

LAKSHAY MANCHANDA

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EDUCATION

University of Petroleum and Energy Studies (UPES), Dehradun
B.Tech, Computer Science (AI & ML)

2024 – 2028
CGPA: 8.46 / 10 | Class XII: 77% | Class X: 96.8%

TECHNICAL SKILLS

Languages: C/C++, Python, Java, TypeScript, JavaScript
Frontend / Mobile: React, Next.js, React Native, Tailwind CSS, Vite
Backend: Node.js, REST APIs
Databases & ORM: MySQL, Prisma ORM, Pinecone (Vector DB)
AI / ML: Google Gemini API, NumPy, Pandas, Scikit-learn, Matplotlib
Tools & Platforms: Git, GitHub, Jira (Agile Kanban), Vercel, Figma

EXPERIENCE

React and React Native Developer Intern (Remote) | Frintt Studio Pvt. Ltd. · [Back&Bone](#) Nov 2025 – Mar 2026

- Engineered useful components of the Back&Bone Website and core screens for Back&Bone, a cross-platform fitness app (React Native 0.73+, Android & iOS), building a modular component library of 20+ reusable UI elements that maintained design consistency across all product surfaces.
- Integrated RESTful APIs for real-time workout and activity tracking; applied Context API state management to eliminate unnecessary re-renders & measurably improve UI responsiveness; contributed concurrently to the production React web platform.

PROJECTS

Ultimate CS Career Architect ([Deployment Link](#))

Next.js · TypeScript · Google Gemini AI · Prisma ORM · PostgreSQL · NextAuth · Vercel

- Built a 3-layer AI resume audit engine using Google Gemini API — ATS compatibility score (0–100), specialization alignment match, and content impact grade (A–F) — with structured JSON generation, automatic retry logic, and PDF export via react-to-pdf.
- Architected a full-stack system: Next.js API routes, Prisma ORM + PostgreSQL, NextAuth session management, drag-and-drop job tracker (5 pipeline stages), AI mock interviews, and AI-generated Mermaid architecture diagrams; deployed on Vercel.

SaralYojna — Government Scheme Discovery Platform ([Deployment Link](#))

React · Node.js · Express · TypeScript · Prisma · Pinecone · Google Generative AI

- Built civic-tech platform with AI-powered eligibility assessment across 500+ Indian government schemes using Pinecone vector embeddings for semantic similarity scoring; secured Node.js/Express API with JWT auth, 2FA, CSRF protection, Redis caching, and Helmet.js security headers.
- Developed Python ETL pipeline (scrape → clean → validate → seed) to normalize scheme dataset; integrated Google Gemini AI chatbot with Hindi/English i18n support, async job queue for email notifications, PWA service worker, and Swagger/OpenAPI documentation.

Dungeon Escape: Cardbound ([Github Link](#))

Java · LibGDX · LWJGL3 · Gradle · LDK · GLSL | Group Project — 6 Members · Agile Kanban (Jira)

- Implemented custom A* pathfinding (F=G+H, Manhattan heuristic, wall-adjacency cost penalty) for organic enemy navigation; built hardware-accelerated FBO lighting via OpenGL blend equations for dynamic radial light falloff across torch, campfire, and lava tile sources.
- Architected JSON DTO save/load system (3 slots) with LibGDX serialization; designed 3-phase boss state machine (Player Turn → Boss Thinking → Boss Turn) with full deck-builder mechanics; wrote GLSL post-processing blur shader for glassmorphism UI; distributed as standalone Gradle JAR.

TaskManager CRUD — Cross-Platform Mobile App ([Github Link](#))

React Native · TypeScript · Context API · AsyncStorage · React Native Reanimated · React Navigation

- Offline-first task management app with full CRUD, priority tiers, time-based reminders, and encrypted AsyncStorage persistence; achieved stable 60 fps animations on Android 5.0+ and iOS 12.0+ via React Native Reanimated.
- Implementing biometric authentication (Face ID / fingerprint) to enhance security for sensitive task data.

LEADERSHIP & ACTIVITIES

Smart India Hackathon (SIH) 2025 — Team Lead

- Led a 6-member cross-functional team to build a multilingual healthcare chatbot (React, Node.js) for Odia-speaking rural communities, improving telemedicine accessibility through symptom-based interaction — competed at national level.

PR Core Member — OPEN Community, UPES

Doc and Scoring Committee Member — PRAYUDH (Debating Club, UPES)

- Part of Organising Committee of National Debating Competition, developing structured argumentation and public speaking skills.